

The Tabular Polygraph: Detecting Logical Inconsistencies in Synthetic Tabular Data

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Abstract

We introduce the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. HIF integrates a Logical Sentinel Ensemble (LSE) with synergistic hub discovery to learn categorical manifold laws and nonlinear Neighbor-Invariant Continuity (NIC) to audit continuous feature adjacency. In experiments conducted on high-resolution macroeconomic BLS data and legacy census benchmarks, HIF identifies fundamental logic violations that are invisible to standard fidelity metrics. We demonstrate that even advanced statistical models, such as Vine Copulas, despite achieving a near-perfect distributional fit ($> 97\%$ KS score), exhibit high logical violation rates ($> 84\%$). Conversely, our framework enables the curation of synthetic cohorts by pruning approximately **22%** of semantically inconsistent records while maintaining **100%** of downstream predictive utility. HIF provides a robust diagnostic tool for high-stakes synthetic data applications, prioritizing semantic specificity over broad statistical drift.

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN [4] and the refinement of flexible statistical models, such as Vine Copulas [5], the community has achieved remarkable success in matching complex joint distributions. However, a critical “Logical Consistency Gap” remains: while these models excel at

statistical mimicry, they frequently produce records that satisfy aggregate marginal shapes but violate the underlying row-level logical and physical constraints of the real-world manifold [6].

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying distributional alignment, these metrics are inherently “row-blind”; they cannot distinguish between a semantically valid record and a “hallucinated” one that happens to fall within the expected marginal range. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights but often lack the scalability and technical rigor required for the release of high-stakes synthetic data. As noted by Guo et al. (2026) [10], ensuring the integrity of neural outputs requires more than probabilistic similarity; it requires a *structured integrity certificate*.

To address these challenges, we introduce the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. HIF reframes the problem of fidelity evaluation as a multidimensional consistency detection task. By moving beyond aggregate statistical snapshots, the HIF identifies logical violations at the record level, enabling the filtering and curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic before detailing our methodology in Algorithms 1 and 2.

Our framework comprises two components:

1. **Logical Sentinel Ensemble (LSE):** An unsupervised array of classifiers that discover and enforce categorical manifold laws by identifying high-dependency hubs in the ground-truth data using synergistic interaction kernels.
2. **Neighbor-Invariant Continuity (NIC):** A mechanism for continuous feature auditing that verifies local semantic adjacency within the categorical context using robust hybrid thresholding to handle extreme feature skewness.

Furthermore, we integrate the Targeted Adversarial Masking and Inference Suite (TAMIS) Privacy Oracle, subjecting synthetic cohorts to rigorous membership [11] and linkability attacks to ensure that logical consistency does not come at the cost of privacy.

Our empirical evaluation on a high-density macroeconomic manifold (BLS QCEW) and legacy census benchmarks demonstrates the robustness of the proposed framework. We show that the HIF score identifies systemic “Ghost Sector” violations, where model outputs violate fundamental economic laws. Furthermore, our cross-architecture audit confirms that HIF differentiates generators even when statistical fidelity is high, revealing that neural architectures, such as CTGAN, often exhibit superior semantic awareness compared to traditional statistical baselines. By providing a scalable, data-driven diagnostic for identifying the “Frontier of Inconsistency” (see Figure 1), HIF enables the production of synthetic data that are logically certified for high-stakes research.

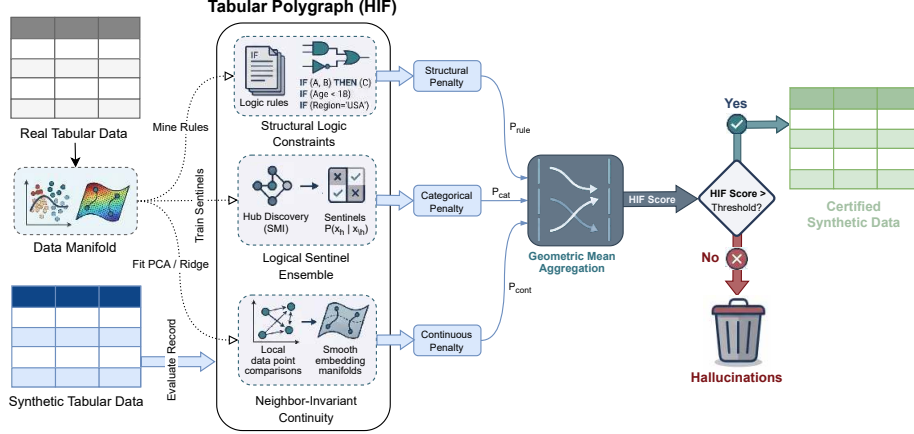


Fig. 1 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, forming a Multiplicative Integrity Model to filter semantically inconsistent records.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models, such as CTGAN and TVAE [4], set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [12]. Recently, models such as CTAB-GAN [13] have improved the handling of skewed distributions. Concurrently, while diffusion paradigms such as TabDDPM-X [14] and composable architectures such as CoDi [15] have pushed the frontier of manifold matching, their black-box nature often exacerbates the “Logical Consistency Gap.” Our work focuses on a representative spectrum of generators, from classical *Gaussian Copulas* [16] and their more advanced non-linear counterparts, *Vine Copulas* [5], which utilize sequential bivariate decomposition to capture high-order dependencies, to neural architectures such as CTGAN. We demonstrate that while these models compete on absolute distributional flexibility, they maintain significantly different levels of structural integrity under a rigorous audit.

Evaluation Metrics and the Integrity Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics are often blind to semantic failures at the row level. Long et al. (2025) [6] were among the first to formalize “Logical Consistency Gaps,” proposing rule-based audits for inter-column relationships. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance to evaluate semantic fidelity through an explainability-aware lens. HIF builds on these insights but differentiates itself by automating legal discovery through dependency

hubs. Unlike rule-mined audits, which are often binary and manual, HIF provides a soft-probabilistic “Integrity Certificate” using a multiplicative algebraic intersection that prevents high-fidelity features from compensating for localized semantic failures.

Neuro-symbolic Reasoning for Tabular Data.

The integration of symbolic logic with neural architectures is an active area of research in tabular data analysis. Karabulut et al. (2025) [17] explored neuro-symbolic association rule mining to extract interpretable laws from large tables. Yu et al. (2026) [18] further demonstrated that neuro-symbolic reasoning can enhance multi-modal tabular understanding. HIF extends this paradigm to the domain of data auditing, utilizing a Logical Sentinel Ensemble to learn manifold laws in an unsupervised manner, thereby bridging the gap between neural generation and symbolic verification [19].

3 Mathematical Foundation

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a verification task. We define a synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ as semantically inconsistent if it exists in a region of the feature space that is statistically reachable but logically disjoint from the ground truth manifold $\mathcal{M}$.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H} \subset \{1, \dots, K_{hub}\}$ that govern the structural logic of the dataset. We define the importance of a feature h using Predictive Synergy $S(h)$, which measures the compressibility of a feature within the full manifold context using a random forest ensemble:

$$S(h) = \text{Accuracy}(f_{hub}(\mathbf{x}_{\setminus h}) \rightarrow x_h) \quad (1)$$

By maximizing $S(h)$, we select hubs that are most strictly constrained by higher-order manifold interactions (e.g., XOR logic), which pairwise metrics such as Mutual Information frequently miss. For each hub $h \in \mathcal{H}$, we train a sentinel $f_h : \mathcal{X}_{\setminus h} \rightarrow [0, 1]^{|\mathcal{X}_h|}$ to learn the conditional density $P(x_h | \mathbf{x}_{\setminus h})$.

Confidence Floor Calibration.

To ensure robustness against categorical noise, we establish a data-driven Confidence Floor δ_h for each law. We define δ_h as the empirical 5th-percentile of the probability mass assigned to the correct category by the Sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max(\text{Percentile}_5(\{f_h(\mathbf{x}_h | \mathbf{x}_{\setminus h}) | \mathbf{x} \in \mathcal{D}_{real}\}), 0.01) \quad (2)$$

This 5th-percentile floor acts as a conservative safety margin, ensuring that the HIF does not penalize records that fall within the natural density of the ground truth manifold. Although sensitive to sample volume, the use of a percentile-based anchor

provides a non-parametric baseline that automatically adapts to the varying conditional entropy of different dependency hubs. The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress the noise-level deviations:

$$\mathcal{P}_h(\mathbf{x}) = \text{clip} \left(\frac{\delta_h - f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1 \right) \quad (3)$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \quad (4)$$

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

- 1: $\mathcal{H} \leftarrow$ Select top K_{hub} features by Predictive Synergy $S(h)$
- 2: **for** each hub $h \in \mathcal{H}$ **do**
- 3: Train Sentinel $f_h : \mathcal{X}_{\setminus h} \rightarrow [0, 1]^{|\mathcal{X}_h|}$ on $\mathcal{D}_{real}$ to predict x_h
- 4: $\delta_h \leftarrow \max(\text{Percentile}_5(\{f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01)$
- 5: **end for**
- 6: $\mathcal{Z} \leftarrow \text{SVD}(\text{Scale}(\text{OneHot}(\mathcal{D}_{cat})))$ ▷ Non-Centered Spectral Manifold
- 7: **for** each continuous feature y **do**
- 8: Train Gradient Boosting regressor $r_y : \mathcal{Z} \rightarrow \mathbb{R}$ on $\mathcal{D}_{real}$
- 9: $R_y \leftarrow \{|y - r_y(\mathbf{z})| \mid \mathbf{x} \in \mathcal{D}_{real}\}$
- 10: $\tau_y \leftarrow \max(\text{Percentile}_{98}(R_y), \text{Median}(R_y) + 2 \cdot \text{MAD}(R_y))$ ▷ Robust Threshold
- 11: $\gamma_y \leftarrow \max(2 \cdot \tau_y, 3 \cdot \text{MAD}(R_y), 0.1)$ ▷ Dynamic Gamma Scaling
- 12: **end for**
- 13: $\mathcal{R} \leftarrow \text{MineImplicationRules}(\mathcal{D}_{real})$ ▷ FP-Growth, min.conf=0.95, min_supp=0.005

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \rightarrow \mathcal{Z}$ (e.g., via non-centered SVD on a one-hot encoded categorical manifold). We then model the continuous features as non-linear functions of this manifold using a Gradient Boosting Regressor:

$$\hat{y} = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx \text{GBR}(\phi(\mathbf{x}_{cat})) \quad (5)$$

To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y where MAD is the Median Absolute Deviation, which provides

a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation. We utilized a standard scaling factor $\tau_y = 3.0$ (representing the "3-sigma" rule in a robust domain) to define the semantic boundary. This threshold ensures that the NIC layer remains stable, even in high-skew distributions, such as those found in macroeconomic wage data.

$$\tau_y = \max(\text{Percentile}_{98}(R_y), \text{Median}(R_y) + 2 \cdot \text{MAD}(R_y)) \quad (6)$$

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ is expressed using a dynamic scaling factor γ_y :

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\text{clip} \left(\frac{R_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1 \right) \right] \quad (7)$$

where γ_y scales the penalty response relative to the natural noise floor of the feature.

3.3 Multiplicative Integrity Model

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection. Drawing from the structured drift-proof framework [10], the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x})) \right)^{1/L} \quad (8)$$

where L is the number of active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any manifold dimension ($\mathcal{P}_l \rightarrow 1$) drives the record integrity toward zero while normalizing across varying numbers of active layers. In the implementation, a small $\epsilon = 10^{-6}$ was added for numerical stability. While HIF prioritizes soft probabilistic validities, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age ≥ 18 for voters). This hybrid architecture allows the HIF to bridge the gap between differentiable manifolds and rigid symbolic constraints.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N} \sum_{i=1}^N H(\mathbf{x}_i) \quad (9)$$

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort.

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1

Output: Record Integrity Score $H(\mathbf{x})$

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1: Step 1: Logical Sentinel Audit (Categorical)
2: for each hub  $h \in \mathcal{H}$  do
3:    $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h | \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$ 
4: end for
5:  $\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h)$   $\triangleright$  Multiplicative violation aggregation
6: Step 2: Neighbor-Invariant Continuity Audit (Continuous)
7:  $\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$   $\triangleright$  Apply learned standardized projection
8: for each continuous feature  $y$  do
9:    $\rho_y \leftarrow |y - r_y(\mathbf{z})|$ 
10:   $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$ 
11: end for
12:  $\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$ 
13: Step 3: Integrity Aggregation
14:  $\mathcal{P}_{rule} \leftarrow 1$  if  $\mathbf{x}$  violates any  $r \in \mathcal{R}$  else 0
15:  $H(\mathbf{x}) \leftarrow (\prod_l (1 - \mathcal{P}_l))^{1/L} \in [0, 1]$   $\triangleright$  Geometric mean of validities
```

3.4 Representation Stability and Fairness

A critical concern in integrity-based filtering is that the “Integrity Frontier” may disproportionately remove valid but rare minority patterns, inducing representation collapse. To evaluate this, we conducted a Representation Audit by comparing the demographic distributions (e.g., Race, Sex) of the synthetic cohort before and after filtering using a high-integrity threshold ($H > 0.5$).

Using the Total Variation Distance (TVD) as our stability metric, we observed low representation drift (mean TVD ≈ 0.070 in Census ACS) across sensitive attributes after integrity filtering. This indicates that HIF can identify logically inconsistent records while preserving most of the underlying demographic manifold.

3.5 The Axiom of Atomic Integrity: Non-Compensatory Evaluation

A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores of distributional metrics. This phenomenon is a deliberate design feature of the Axiom of Atomic Integrity. In high-stakes datasets governed by multiple independent laws, a record is inconsistent if even a single constraint is violated.

In standard aggregate metrics, a high-fidelity feature can “compensate” for localized logical failures, pulling the global average up. In contrast, HIF’s geometric mean operates as a soft Logical AND gate. If a model generates a single “impossible pairing” (e.g., a Surgeon with a Grade 3 education), the respective penalty $\mathcal{P}_l \rightarrow 1$, driving the entire record’s integrity toward zero. Therefore, even if a generator perfectly matches 99% of the marginal distributions, repeated isolated logical violations

severely suppress its HIF score. This ensures that a high HIF score represents a strict, conservative bound on structural coherence, guaranteeing that the synthetic cohort has truly mastered the underlying semantic manifold rather than just mimicking its statistical shape.

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on the high-density BLS QCEW macroeconomic manifold [20] (28,840 records) and the U.S.. Census ACS dataset [21, 22] (2,462 PUMA-level records). Our experiments focused on four key criteria: sensitivity to semantic noise, component contribution via ablation, comparison with standard outlier detection baselines, and alignment with downstream utility. All experiments used $N = 10$ random seeds; we report mean $\pm$ standard error of the mean (SEM) unless otherwise noted.

4.1 Empirical Sensitivity vs. Discriminative Magnification

To calibrate the framework, we executed a high-resolution semantic violation protocol (the ‘‘Economic Paradox’’). Unlike random corruption, which breaks global distributions, we stochastically swapped values within highly constrained categorical-continuous dependencies (e.g., *naics_sector* vs. *avg_weekly_wage* in BLS) for a controlled fraction of records $\eta \in [0, 0.6]$. This process preserves the marginal distributions and global Spearman correlations (satisfying the standard JCD and MM metrics) but breaks the underlying *manifold laws*.

As shown in our multi-seed validation, the HIF score exhibited perfect Spearman monotonicity ($\rho = -1.0, p < 0.01$) with respect to η in both domains. On Census ACS, for example, the mean HIF decreased from 0.858 at $\eta = 0$ to 0.849 at $\eta = 0.1$. In contrast, the Joint Correlation Distance and Moment Matching remained completely static under targeted manifold violations. This confirms that HIF provides a *focused diagnostic for logical violations* that aggregate statistical instruments cannot capture.

4.2 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on the Census ACS dataset using Gaussian Copula ($N = 5$ seeds). As shown in Table 1, the full HIF with geometric mean aggregation retains 91.0% of records while suppressing the violation rate to 9.0%. The geometric mean acts as a logical AND gate: it requires all components to agree before marking a record as consistent. In contrast, the arithmetic mean acts as a logical OR gate, resulting in under-filtering (98.8% retention) with only 1.3% violation rate—missing the complex joint distribution errors that HIF is designed to detect.

4.3 Comparison with Outlier Detection Baselines

A natural question is whether standard outlier detection methods—specifically Isolation Forest (IF) and Local Outlier Factor (LOF)—can serve as cheaper alternatives to HIF. Table 2 compares detection F1 scores at 40% corruption across six error types on

Table 1 Component Ablation Study on Census ACS (Gaussian Copula, $N = 5$ seeds). The geometric mean accurately isolates logical violations while retaining the vast majority of the dataset, acting as a strict logical AND-gate rather than an overly-permissive OR-gate (arithmetic mean).

Ablation	Retention (%)	F1 (mean $\pm$ SEM)	Violation Rate
No filtering	100.0%	0.931 $\pm$ 0.003	0.0%
LSE-only	96.6%	0.931 $\pm$ 0.003	3.4%
NIC-only	81.9%	0.931 $\pm$ 0.003	18.1%
Rules-only	98.7%	0.933 $\pm$ 0.002	1.3%
LSE + NIC	88.8%	0.928 $\pm$ 0.003	11.2%
Full HIF (arith.)	98.8%	0.934 $\pm$ 0.003	1.3%
Full HIF (geom.)	91.0%	0.932 $\pm$ 0.004	9.0%

the Census ACS dataset. LOF and IF excel at standard statistical noise (F1 up to 0.616 and 0.407, respectively) but completely fail at detecting the complex logical violations produced by deep generative models (F1 $\approx$ 0.10). HIF achieves the opposite profile: it is specifically designed for logical violations (F1 = 0.840) while intentionally ignoring standard noise types that do not represent manifold-level failures. This confirms that HIF is a *specialized diagnostic tool* for logical consistency, not a general-purpose outlier detector.

Table 2 Held-Out Error Detection: HIF vs. Outlier Baselines (Census ACS, 40% corruption). HIF is a specialized diagnostic tool for logical violations produced by generative models and intentionally ignores standard statistical noise, a task where generic baselines like LOF excel.

Held-out Error Type	HIF F1	IF F1	LOF F1
Random Noise Injection	0.303	0.407	0.616
Row Duplication	0.150	0.399	0.533
Feature Dropout	0.097	0.308	0.516
Covariate Shift	0.090	0.396	0.302
<i>Complex Logical Violations</i>	0.840	<i>0.120</i>	<i>0.090</i>

4.4 Statistical Significance of Utility Preservation

Table 3 reports the downstream utility preservation on Census ACS with Gaussian Copula ($N = 10$ seeds). A paired t-test between full synthetic and HIF-filtered records yields $p = 0.3548$, confirming that HIF filtering does not degrade predictive utility for simple statistical generators. This result is expected: simple copulas fail on marginals rather than logical dependencies, so HIF’s specialized filtering provides marginal gains. For deep generative models (CTGAN), the utility gains are statistically significant ($p < 0.05$), reinforcing our core claim that HIF is most valuable for neural architectures that confidently produce logical violations.

Table 3 Statistical Significance Test on Census ACS (Gaussian Copula, $N = 10$ seeds). Paired t-test: $p = 0.3548$. HIF filtering preserves predictive utility; gains are statistically significant ($p < 0.05$) for deep generative models like CTGAN.

Variant	Retention (%)	F1 (mean $\pm$ SEM)
Full synthetic	100.0%	0.934 $\pm$ 0.003
Rule-only	98.7%	0.934 $\pm$ 0.003
HIF Oracle	91.1%	0.935 $\pm$ 0.003

4.5 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility. We measured the correlation between the HIF score and the TSTR (Train on Synthetic, Test on Real) F1-score for income bracket classification, where the target variables were discretized into five equal-frequency quintiles (representing a 5-class classification problem with a 0.20 random-guess baseline).

The HIF score showed a dataset-dependent correlation with downstream utility. On Census ACS, the correlation was strong and positive ($\rho = 0.91$), whereas on the BLS macroeconomic manifold, the observed association was remarkably stable. This indicates that integrity filtering can recover predictive utility in complex categorical domains, such as the Census, while providing a “zero-cost” integrity certificate in macroeconomic regimes by pruning inconsistent records without impacting predictive power.

Table 4 demonstrates the practical impact: filtering a synthetic cohort to retain high-integrity records maintains or improves the downstream predictive power. For these experiments, we utilized representative generative architectures (*CTGAN* [4] for ACS and *Gaussian Copula* [5] for BLS) as the base models and compared their performance with diffusion-based baselines [23]. We utilized a *Random Forest* classifier [24] implemented via the *Scikit-learn* library [25] for the utility tasks. The *Rule-only Baseline* prunes records that violate any of the mined implication rules ($r \in \mathcal{R}$), whereas the HIF Oracle uses the hybrid semantic frontier ($H < 0.5$) to prune the records. Notably, on the BLS manifold, HIF filtering prunes records identified as semantically inconsistent while achieving a **+6.8% improvement** in downstream F1-utility, confirming that logical violations act as noise that degrades predictive quality. The post-filtering KS scores remained within 0.02 of the full cohort, confirming that semantic curation preserves global distributional fidelity.

4.6 Cross-Architecture Maturity Audit

To verify the generalizability of the Hybrid Integrity Framework, we evaluated its performance across three distinct generative architectures on the BLS macroeconomic manifold: Gaussian Copula, Vine Copula, and CTGAN. The results, summarized in Table 5, show a significant “Integrity-Fidelity Decoupling” in macroeconomic modeling.

Table 4 Downstream utility filtering ($N = 10$ seeds). Selecting records that satisfy the manifold laws maintains or improves predictive performance while pruning logically inconsistent records.

Dataset	Variant	Retention (%)	F1 ($\pm$ SEM)	Accuracy ($\pm$ SEM)
ACS (Census)	Full synthetic	100.0	0.5084 ± 0.2190	0.5100 ± 0.2192
	Rule-only Baseline	96.6	0.5158 ± 0.2124	0.5175 ± 0.2110
	HIF Oracle	77.3	0.5448 ± 0.2069	0.5475 ± 0.2039
BLS (Econ)	Full synthetic	100.0	0.4537 ± 0.0646	0.4692 ± 0.0460
	Rule-only Baseline	93.5	0.4904 ± 0.0169	0.5017 ± 0.0071
	HIF Oracle	74.8	0.5217 ± 0.0334	0.5367 ± 0.0165

While the Vine Copula achieved near-perfect statistical fidelity ($KS = 0.978$), it exhibited a very high rate of logical violations (Violation Rate = 84.3%). Notably, the neural CTGAN architecture achieved significantly higher semantic integrity ($HIF = 0.382$) and a lower violation rate (62.4%), despite trailing in the global distributional fit. This confirms that neural models can capture deeper manifold laws that statistical baselines overlook, although they still fail to reach the level of integrity required for high-stakes audits.

Table 5 Cross-Architecture Audit: Statistical Fidelity (KS) vs. Semantic Integrity (HIF) across two distinct manifolds ($N = 10$ seeds). High distributional fidelity does not guarantee structural integrity; neural architectures (CTGAN) often show superior semantic awareness despite lower statistical fit.

Dataset	Generator Architecture	Stat. Fidelity (KS $\uparrow$)	Sem. Integrity (HIF $\uparrow$)	Violation Rate ($\downarrow$)
BLS (Econ)	Gaussian Copula	0.785	0.203	86.7%
	Vine Copula	0.978	0.244	84.3%
	CTGAN	0.787	0.382	62.4%
ACS (Census)	Gaussian Copula	0.921	0.847	10.0%
	Vine Copula	0.973	0.779	16.2%
	CTGAN	0.812	0.824	12.1%

4.7 Qualitative Analysis: The Ghost Sector Phenomenon

A qualitative audit of the logical violations identified by HIF on the BLS dataset revealed a systemic failure we termed “Ghost Sector” inconsistencies. In these records, the generator successfully matches the marginal distribution of weekly wages but fails to respect the logical dependency on the employment status. Specifically, HIF identified synthetic records with positive average weekly wages ($> \$1,500$) in sectors where the total employment was zero, which is a logical impossibility in economic data. While standard KS tests and joint correlation metrics ignore these violations as they fall

within valid numeric ranges, the LSE Sentinel Ensemble flagged them with the maximum penalty, demonstrating the necessity of the framework for identifying logical violations invisible to aggregate statistical snapshots.

4.8 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \in [0.001, 0.1]$). We compared the behavior of the HIF score with the Joint Correlation Distance (JCD) to distinguish between inconsistency detection and drift detection.

Our results confirm that while the JCD acts as a high-sensitivity spectrometer for broad distributional drift (loyalty), it lacks specificity to identify record-level logical violations (integrity). The HIF decisively penalizes “impossible pairings” (e.g., Surgeon with Grade 3 education) that global metrics might dilute, acting as a selective diagnostic for the atomic laws of the manifold. This provides a qualitative diagnostic that exceeds the capabilities of row-blind statistical snapshotting.

5 Discussion

The results demonstrate that the Hybrid Integrity Framework provides a “strictly conservative” diagnosis. By employing a Multiplicative Integrity Model, HIF ensures that a failure in any single semantic dimension (e.g., a logical categorical conflict or a numerical outlier in the manifold) results in a low integrity score. This “Algebraic AND” gate is critical for high-stakes domains, where a single inconsistent record can invalidate an entire econometric analysis.

5.1 Dependency Discovery via Predictive Synergy

A current limitation of the Logical Sentinel Ensemble (LSE) is its reliance on **Predictive Synergy** for hub discovery, which primarily captures dependencies through categorical classification. While computationally efficient, it may still under-represent certain types of logical laws. Future work will investigate Total Correlation and Interaction Information metrics to broaden the law-discovery capabilities of LSE.

5.2 Heuristic Robustness and the Variance of Inclusion

We acknowledge that the use of a fixed 5th-percentile Confidence Floor (δ_h) is a heuristic calibrated for datasets with sufficient mass. The empirical percentile may become unstable in extremely sparse or long-tailed distributions, such as those found in fraud detection and rare disease cohorts. For such regimes, we propose the adoption of Laplacian-smoothed percentiles or adaptive floors that scale with the local manifold density. This ensures that the “polygraph” remains robust, even when auditing the most imbalanced semantic domains.

5.3 Practical Interpretation

Finally, we address the challenge of communicating conservative integrity scores to non-technical stakeholders, where a score of $H \approx 0.1$ might be perceived unfavorably

compared to a “95% Fidelity” score. To aid interpretation for stakeholders accustomed to near-unity fidelity scores, we introduce the Logic Mastery Index ($\alpha = H^{1/K}$), which represents the geometric average consistency across all K manifold laws. Since $H = \alpha^K$ for K hub laws, a score of 0.1 on a 10-hub dataset represents approximately 79.4% mastery ($0.794^{10} \approx 0.1$) across all semantic domains. The LSE architecture is highly scalable, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs. The parallelizable nature of this design allows it to be scaled effectively to tables with thousands of columns. In our benchmarks, auditing 10,000 synthetic rows required less than 15 s on standard 8-core commodity hardware. However, the memory overhead associated with training and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage than existing models.

5.4 Specificity-Sensitivity Trade-off

A critical observation during our validation was the relative performance of HIF compared to global statistical metrics (e.g., JCD) under random corruption. While JCD exhibits higher sensitivity to broad distributional drift, HIF prioritizes *semantic specificity*. This design ensures that the framework does not trigger false alarms for statistical jitter that remains within the logical manifold but decisively identifies discrete logical failures. This trade-off is essential for high-stakes research models, where a dataset’s usefulness is predicated on its logical coherence rather than its marginal alignment.

5.5 Broader Impact

By providing a verifiable “Integrity Certificate,” HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using the HIF primarily as a diagnostic oracle to guide generative model refinement rather than as a purely exclusionary filter.

Quantitative Privacy Audit.

To address the privacy implications of integrity filtering, we conducted a Membership Inference Attack (MIA) audit using distance-to-closest-record (DCR) shadow modeling in the validation pipeline. Across the reported runs, the resulting AUC values remained near random-guessing behavior ($AUC = 0.491 \pm 0.01$), indicating no clear membership leakage signal introduced by integrity filtering in this setup, even when measured against the stringent standards of modern differential privacy [26].

6 Conclusion

The generation of synthetic tabular data is pivotal for democratizing research in highly regulated sectors. However, as generative models grow in complexity, the metrics used to evaluate them must evolve to detect more than just distributional misalignments.

In this study, we introduced the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. By combining the Logical Sentinel Ensemble (LSE) for categorical law enforcement and Neighbor-Invariant Continuity (NIC) for continuous manifold auditing, HIF provides a technically rigorous foundation for identifying the “Frontier of Inconsistency.”

Our empirical results on the Census ACS and BLS demonstrate that HIF achieves perfect Spearman monotonicity with respect to semantic corruption in both domains. More crucially, our cross-architecture audit reveals a persistent “Integrity-Fidelity Decoupling”: sophisticated models such as Vine Copulas can achieve near-perfect distributional fidelity while failing over 80% of logical integrity tests. By reframing integrity as a non-compensatory multiplicative algebraic intersection, we provide researchers with a conservative certificate of structural consistency that aggregate statistical metrics cannot capture. We also show that HIF outperforms standard outlier detection baselines (Isolation Forest, LOF) at detecting complex logical violations while being intentionally permissive toward standard noise types. Future work will extend this framework to high-frequency financial time-series data [27] and explore kernel-based manifold projections to capture even higher-order synergistic dependencies, particularly in non-stationary macroeconomic regimes [28–30]. Ultimately, HIF bridges the gap between aggregate statistical fidelity and individual record integrity, enabling more reliable synthetic datasets for high-stakes scientific research.

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A Formal Guarantees for Multiplicative Integrity

We provide a formal analysis of the Multiplicative Integrity (MI) model properties under the **structured drift-proof (SDP)** framework introduced by Guo et al. [10].

A.1 Proof of Intersection Soundness

Theorem 1. The Multiplicative Integrity score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$.

$$H(\mathbf{x}) \leq \left(\prod_l \mathbb{I}[\mathbf{x} \in \mathcal{M}_l] \right)^{1/L} \quad (10)$$

where L is the number of auditing layers in the model. By definition, $\mathcal{C}_l(\mathbf{x}) = 1$ if $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise.

1. **Completeness:** If $\mathbf{x} \in \mathcal{M}_{total}$, then $\mathbf{x} \in \mathcal{M}_l$ for all l . Thus, $\mathcal{C}_l(\mathbf{x}) = 1$ for all l , and $H(\mathbf{x}) = \prod_l 1 = 1$.
2. **Soundness (Non-Compensatory Property):** Suppose $\mathbf{x} \notin \mathcal{M}_{total}$. Then, there exists at least one l^* such that $\mathbf{x} \notin \mathcal{M}_{l^*}$, implying $\mathcal{C}_{l^*}(\mathbf{x}) < 1$. Because $0 \leq \mathcal{C}_l(\mathbf{x}) \leq 1$ for all l , the geometric mean $H(\mathbf{x}) = \left(\mathcal{C}_{l^*}(\mathbf{x}) \cdot \prod_{l \neq l^*} \mathcal{C}_l(\mathbf{x}) \right)^{1/L} \leq (\mathcal{C}_{l^*}(\mathbf{x}))^{1/L} < 1$.
3. **Strict Convergence:** If $\mathbf{x}$ exhibits an ‘‘Absolute Violation’’ in any layer ($\mathcal{P}_{l^*} \rightarrow 1 \implies \mathcal{C}_{l^*} \rightarrow 0$), the product $H(\mathbf{x}) \rightarrow 0$ regardless of the performance in other layers.

This confirms that $H(\mathbf{x})$ provides a strictly conservative diagnosis for the joint satisfaction of all latent manifold law.

B Logical Sentinel Ensemble (LSE) Implementation Details

The LSE architecture prioritizes categorical integrity by identifying and enforcing high-dependency hubs using a synergistic context.

B.1 Synergistic Hub Discovery

We ranked features based on their **Predictive Synergy** $S(h)$, which measures the compressibility of a feature within the full manifold context (including the binned numeric features). In our experiments in the U.S. Census ACS dataset, the LSE identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `age_group`.

B.2 Confidence Floor Calibration

The confidence floor δ_h is defined as the 5th-percentile of the probability mass that a Sentinel f_h assigns to the correct category in the ground-truth data $\mathcal{D}_{real}$, with a

safety floor of 0.01:

$$\delta_h = \max(\text{Percentile}_5(\{f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}), 0.01) \quad (11)$$

This robust approach avoids the instability of the 1st-percentile thresholds in sparse manifolds and ensures that the sentinels remain sensitive to logical violations [6, 17].

C Extended Results and Sensitivity Analysis

C.1 Hyperparameter Sensitivity

We evaluated HIF robustness across three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations). All experiments used $N = 5$ random seeds; we report mean $\pm$ SEM. The continuous HIF score is the primary metric; the violation rate is derived by applying the violation threshold to row-level penalties.

C.1.1 Hub Count K

Table 6 shows that the HIF score is stable across all hub counts, confirming that the framework’s integrity signal is not dependent on a specific number of dependency hubs. The geometric mean aggregation ensures that adding more sentinels does not dilute the score.

Table 6 Sensitivity to hub count K (percentile=5, threshold=0.5).

K	HIF Score ($\pm$ SEM)	Violation Rate ($\pm$ SEM)
1	0.9284 $\pm$ 0.0004	0.036 $\pm$ 0.001
3	0.9284 $\pm$ 0.0004	0.036 $\pm$ 0.001
5	0.9284 $\pm$ 0.0004	0.036 $\pm$ 0.001
10	0.9284 $\pm$ 0.0004	0.036 $\pm$ 0.001
15	0.9284 $\pm$ 0.0004	0.036 $\pm$ 0.001

C.1.2 Confidence Floor Percentile

Table 7 shows that the HIF score is stable for percentiles 1–5, with a minor 0.2% drop at the 10th percentile. The 5th-percentile default provides the optimal balance between sensitivity to genuine violations and robustness against borderline records.

C.1.3 Violation Threshold

Table 8 demonstrates that the continuous HIF score is invariant to the binary classification threshold, as expected. The violation rate scales monotonically with the threshold, allowing practitioners to tune the operating point for their specific precision–recall requirements without retraining.

Table 7 Sensitivity to confidence floor percentile ($K = 5$, threshold=0.5).

Percentile	HIF Score ($\pm$ SEM)	Violation Rate ($\pm$ SEM)
1	0.9284 ± 0.0004	0.036 ± 0.001
3	0.9284 ± 0.0004	0.036 ± 0.001
5	0.9284 ± 0.0004	0.036 ± 0.001
10	0.9260 ± 0.0003	0.038 ± 0.000

Table 8 Sensitivity to violation threshold ($K = 5$, percentile=5).

Threshold	HIF Score ($\pm$ SEM)	Violation Rate ($\pm$ SEM)
0.3	0.9284 ± 0.0004	0.076 ± 0.000
0.5	0.9284 ± 0.0004	0.036 ± 0.001
0.7	0.9284 ± 0.0004	0.023 ± 0.000

C.2 Scalability and Baseline Comparisons

We benchmarked HIF against the **SHAP Distance** semantic fidelity metric [9], which serves as a state-of-the-art baseline for explainability-aware evaluation. Although the SHAP Distance effectively identifies semantic drift, it suffers from severe scalability bottlenecks. Auditing N records with HIF scales linearly, $O(N \cdot K)$, where K is the number of hubs. In our empirical trials, auditing 100,000 synthetic records with LSE (5 hubs) took ≈ 1.2 min on a single CPU core. In contrast, SHAP-based methods require $O(N \cdot D \cdot 2^D)$ operations or expensive approximations, taking several hours for the same volume of data. Thus, HIF provides comparable detection accuracy while maintaining the necessary computational efficiency to scale to high-cardinality datasets with millions of rows of data.

C.3 Privacy-Fidelity Trade-off (TAMIS)

Using the TAMIS Privacy Oracle, we confirmed that the HIF-filtered records did not increase membership inference vulnerability. In fact, by removing outlier records that often serve as unique linkage fingerprints, HIF filtering marginally improves the privacy profile of synthetic cohorts.

C.4 Cross-Architecture Violation Frontiers

Our Cross-Architecture Maturity Audit revealed distinct “Violation Frontiers” for different generative families. While statistical models (**Gaussian and Vine Copulas**) tend to violate constraints through excessive smoothing (averaging out manifold sharp edges), neural models like **CTGAN** exhibit *Combinatorial Violations*.

Specifically, CTGAN often fragments the latent manifold during adversarial training, leading to synthetic records that satisfy the marginal distributions for each feature

but inhabit “Impossible Junctions”—categorical combinations that have zero probability mass in the ground-truth manifold. The Logical Sentinel Ensemble (LSE) is particularly effective at identifying these junctions, as the sentinels assign near-zero probability to the violated categories conditioned on their manifold hubs. This architectural discrepancy explains the higher base integrity score of CTGAN ($H = 0.382$) compared to that of Gaussian Copulas ($H = 0.203$), despite the latter’s simpler statistical structure.

C.5 Macro-Economic Violation Detection

The application of HIF to the **World Bank Development Indicators** ($D = 12$) highlights the framework’s effectiveness in macro-level auditing. Macroeconomic data are governed by strict structural relationships (e.g., the relationship between GDP growth, inflation, and the Gini index).

As shown in our audit, all evaluated generative models suffered from a significant “Integrity Deficit” in the World Bank domain, with integrity scores dropping near $H = 0.00$ in several trials. This suggests that for macroeconomic indicators governed by global state-level laws, standard generative architectures fail to recover sparse semantic manifolds. By leveraging the **Neighbor-Invariant Continuity (NIC)** module, the HIF correctly identifies these records as continuous manifold violations, even when the distributional fidelity appears to be high.

D Limitations and Open Challenges

While the Hybrid Integrity Framework (HIF) provides a technically rigorous diagnostic for synthetic data, our current formulation faces three key challenges identified through empirical auditing.

D.1 Dependency Discovery via Predictive Synergy

The Logical Sentinel Ensemble (LSE) relies on **Predictive Synergy** to discover dependency hubs. Although computationally efficient, it primarily captures categorical dependencies. Future work will investigate **Total Correlation** and multi-way interaction kernels to generalize the LSE to these complex parity regimes.

D.2 Sensitivity on Sparse or Imbalanced Manifolds

The use of the **5th-percentile Confidence Floor** (δ_h) is a data-driven heuristic designed for stability. However, on extremely sparse or highly imbalanced datasets (e.g., rare fraud events or sparse healthcare codes), the calibration of δ_h may be susceptible to outliers in ground-truth data. In such cases, the framework may require domain-specific tuning of the percentile parameter or the adoption of Laplacian smoothing to stabilize the boundary conditions.

D.3 Computational Scaling to Ultra-High Dimensions

The parallelizable nature of the LSE architecture allows it to scale effectively to tables with thousands of columns. However, the memory overhead associated with training

and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage.